Stochastics – KVB

Random variable:

Let $(\Omega, \mathcal{F}, P)$ be a probability space. A map $X: \Omega \to \mathbb{R}$ is called a random variable if

$$\{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{F} \text{ for all } A \in \mathcal{B}(\mathbb{R})$$

eg:

Let X be a random variable on a discrete probability space (Ω, P), i.e. a map $X: \Omega \to \mathbb{R}$.
Then, for any $A \subseteq \mathbb{R}$,

$$\{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{P}(\Omega),$$

where X is a random Variable on $(\Omega, \mathcal{P}(\Omega), P)$

For a random variable X define

$$P_X(A) = P(X \in A) = P(X^{-1}(A)) = P(\{\omega \in \Omega : X(\omega) \in A\})$$

for $A \in \mathcal{B}(\mathbb{R})$. Then, $(\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X)$ is a probability space

Distribution function:

For a random variable X, then function

$$F_X : \mathbb{R} \to [0,1], \; y \mapsto P(X \leq y) = P(X \in (-\infty, y])$$

is called the distribution function of X

eg: (Bernoulli Distribution):

A Bernoulli distributed random variable X with parameter $p \in (0,1)$, i.e. $P(X=1) = p$ and $P(X=0) = 1-p$, has the distribution:

$$F_X(y) = P(X \leq y) = \begin{cases} 0 & , y < 0 \\ 1-p & , y \in [0,1) \\ 1 & , y \geq 1 \end{cases}$$

Stochastics -KVB-

3.15.

Let F_X be the distribution of a random variable X.

Then,

(i) F_X is non-decreasing, i.e., $F_X(y_1) \leq F_X(y_2)$ for $y_1 \leq y_2$, $y_1, y_2 \in \mathbb{R}$

(ii) $\lim_{y \to -\infty} F_X(y) = 0$ and $\lim_{y \to \infty} F_X(y) = 1$

(iii) F_X is right-continuous, i.e.

$$\lim_{z \to y^+} F_X(z) = F_X(y) \text{ for } y \in \mathbb{R}$$

3.16.

For $y \in \mathbb{R}$ let $F_X(y^-) = \lim_{z \to y^-} F_X(z)$. Then,

$$F_X(y^-) = P(X < y)$$

Whence

$$P(X = y) = P(X \leq y) - P(X < y) = F_X(y) - F_X(y^-)$$

For $a, b \in \mathbb{R}$ with $a < b$,

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

$$P(a \leq X \leq b) = F_X(b) - F_X(a^-)$$

$$P(a < X < b) = F_X(b^-) - F_X(a)$$

$$P(a \leq X < b) = F_X(b^-) - F_X(a^-)$$

Stochastics - IK/VB

→ 3.17: (distribution Function)

For each $F: \mathbb{R} \to [0,1]$ satisfying (i)-(iii) in 3.15 there exists a probability space $(\Omega, \mathcal{F}, P)$ with a random variable X such that

$$P(X \leq y) = F(y) \quad \text{for } y \in \mathbb{R}$$

→ 3.18:

A function $f: \mathbb{R} \to \mathbb{R}$ is called a density if $f(x) \geq 0$ for $x \in \mathbb{R}$ and

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

→ 3.19:

For a density $f: \mathbb{R} \to \mathbb{R}$,

$$F: \mathbb{R} \ni y \mapsto F(y) = \int_{-\infty}^{y} f(x)\,dx$$

is a distribution Function

→ eg: (Uniform distribution):

For $a, b \in \mathbb{R}$ with $a < b$ let

$$F: \mathbb{R} \to [0,1], \quad F(y) = \begin{cases} 0, & y < a \\ \frac{y-a}{b-a}, & a \leq y \leq b \\ 1, & y > b \end{cases}$$

Since F is non-decreasing, continuous and $\lim_{y \to -\infty} F(y) = 0$ and $\lim_{y \to +\infty} F(y) = 1$,

F is a distribution function. Its density $f(x) = F'(x) = \begin{cases} 0, & x < a \\ \frac{1}{b-a}, & a \leq x \leq b \\ 0, & x > b \end{cases}$

A random variable with distribution F is uniformly distributed on $[a,b]$

→ eg: (Pareto Distribution)

For $\alpha > 0$, def

$$f: \mathbb{R} \to \mathbb{R} \;/\; f(x) = \begin{cases} 0 & , x < 1 \\ \alpha x^{-\alpha-1} & , x \geq 1 \end{cases}$$

Since $f(x) \geq 0$ for $x \geq 0$ and

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{1}^{\infty} \alpha x^{-\alpha-1}\,dx = \left[-x^{-\alpha}\right]_1^{\infty}$$

$$= 0 - (-1) = 1$$

f is a density. For $y < 1$

$$\int_{-\infty}^{y} f(x)\,dx = 0$$

while for $y \geq 1$,

$$\int_{-\infty}^{y} f(x)\,dx = \int_{1}^{y} \alpha x^{-\alpha-1}\,dx = \left[-x^{-\alpha}\right]_1^{y} = 1 - y^{-\alpha}$$

Thus, f has the distribution function

$$F(y) = \begin{cases} 0 & , y < 1 \\ 1 - y^{-\alpha} & , y \geq 1 \end{cases}$$

Stochastics - KVO

→ eg: (Exponential distribution):

For $\lambda > 0$ define $f: \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} 0 & , x < 0 \\ \lambda e^{-\lambda x} & , x \geq 0 \end{cases}$$

Distribution Function:

$$F(y) = \begin{cases} 0 & , y < 0 \\ 1 - e^{-\lambda y} & , y \geq 0 \end{cases}$$

→ eg: (Normal Distribution):

Let the function $\varphi: \mathbb{R} \to \mathbb{R}$ be given by

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad \text{for } x \in \mathbb{R}.$$

Density $\varphi(x)$.

Distribution Function:

$$\Phi : \mathbb{R} \to [0,1], \quad y \mapsto \Phi(y) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{y} e^{-\frac{x^2}{2}} dx$$

For $\mu \in \mathbb{R}$ and $\sigma \in (0,\infty)$ one can define the random variable $X = \mu + \sigma N$, which has the distribution function,

$$F_X(y) = P(X \leq y) = P(\mu + \sigma N \leq y) = P(N \leq \frac{y-\mu}{\sigma}) = \Phi(\frac{y-\mu}{\sigma})$$

for $y \in \mathbb{R}$ and the density

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{\sigma^2}} \quad \text{for } x \in \mathbb{R}$$

↪ Gaussian distribution with parameters μ & σ.

Stochastics - KVB

→ Independence of random variables

Random variables $X_1, \dots, X_n$ are called independent if

$$P(X_1 \in A_1, \dots, X_n \in A_n) = \prod_{i=1}^{n} P(X_i \in A_i)$$

for all $A_1, \dots, A_n \in \mathcal{B}(\mathbb{R})$

→ 3.27:

Let $X_1, \dots, X_n$ be random variables.

1) Then, $X_1, \dots, X_n$ are independent if and only if

$$P(X_1 \leq y_1, \dots, X_n \leq y_n) = \prod_{i=1}^{n} P(X_i \leq y_i)$$

for all $y_1, \dots, y_n \in \mathbb{R}$

2) Let $X_1(\Omega), \dots, X_n(\Omega)$ be at most countable. Then $X_1, \dots, X_n$ are independent if and only if

$$P(X_1 = y_1, \dots, X_n = y_n) = \prod_{i=1}^{n} P(X_i = y_i) \text{ for all } y_1 \in X_1(\Omega), \dots$$

→ eg:

Let $\Omega = \{1, \dots, 6\}^2$ and $P(A) = \frac{|A|}{36}$ for $A \subseteq \Omega$. Define $X_1 : \Omega \ni (\omega_1, \omega_2) \mapsto \omega_1$

& $X_2 : \Omega \ni (\omega_1, \omega_2) \mapsto \omega_2$

For $i_1, i_2 \in \{1, \dots, 6\}$,

$$P(X_1 = i_1, X_2 = i_2) = P(\{(i_1, i_2)\}) = \frac{1}{36}$$

$$P(X_1 = i_1) = \frac{1}{6}, \quad P(X_2 = i_2) = \frac{1}{6}$$

$$P(X_1 = i_1)\, P(X_2 = i_2) = \frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36} = P(X_1 = i_1, X_2 = i_2)$$

↳ independent!

Stochastics - KW 6-6

Let $X_1, \dots, X_n$ be random variables. We call

$$P((X_1, \dots, X_n) \in A) = P(\{\omega \in \Omega : (X_1(\omega), \dots X_n(\omega)) \in A\})$$

for $A \in \mathcal{B}(\mathbb{R}^n)$ the distribution of $X_1, \dots, X_n$ and

$$F_{X_1, \dots, X_n}(y_1, \dots y_n) = P(X_1 \leq y_1, \dots, X_n \leq y_n)$$

for $y_1, \dots, y_n \in \mathbb{R}$ the distribution function of $X_1, \dots, X_n$.
For $i \in \{1, \dots, n\}$ we denote the distribution of X_i as marginal distribution.

Stochastics - Ex 5

(2)

Let X be a Poisson distributed random variable with parameter $\lambda > 0$. Given X, let Y be a binomially distributed random variable with parameters X and $p \in [0,1]$, i.e. for any $x, y = 0, 1, \ldots$ it holds

$$P(X = x) = e^{-\lambda} \frac{\lambda^x}{x!} \quad \text{and} \quad P(Y = y \mid X = x) = \binom{x}{y} p^y (1-p)^{x-y}$$

Show that Y is also Poisson distributed with a parameter $\mu \geq 0$

$$P(Y = y) = \sum_{k=0}^{\infty} P(Y = y \mid X = k) P(X = k)$$

$$= \sum_{k=0}^{\infty} \binom{k}{y} p^y (1-p)^{k-y} \cdot e^{-\lambda} \cdot \frac{\lambda^k}{k!}$$

$$= \sum_{k=0}^{\infty} \frac{k!}{y!(k-y)!} p^y (1-p)^{k-y} \cdot e^{-\lambda} \frac{\lambda^k}{k!}$$

$= 0$ if $y > x$

$k = y$

$s = y + k$

$$= \sum_{k=y}^{\infty} \frac{1}{y!(k-y)!} p^y (1-p)^{k-y} e^{-\lambda} \frac{\lambda^k}{k!}$$

$$\Leftrightarrow \frac{(\lambda p)^y}{y!} e^{-\lambda} \sum_{k=y}^{\infty} \frac{1}{(k-y)!} (1-p)^{k-y} \lambda^{x-y}$$

$$\Leftrightarrow \frac{(\lambda p)^y}{y!} e^{-\lambda} \sum_{x=y}^{\infty} \frac{((1-p)\lambda)^{x-y}}{(x-y)!}$$

$\hookrightarrow x - y = 0$

$$\Leftrightarrow \frac{(\lambda p)^y}{y!} e^{-\lambda} \sum_{s=0}^{\infty} \frac{((1-p)\lambda)^s}{s!} = \frac{(\lambda p)^y}{y!} e^{-\lambda} e^{(1-p)\lambda}$$

$$= \frac{(\lambda p)^y}{y!} e^{-p\lambda}, \quad y = 0, 1, 2, \ldots$$

$\Rightarrow Y$ is poisson

$$\sum_{k=0}^{\infty} \frac{x^k}{k!} = e^x$$

Stochastics - KVB - 6x3

(3)

Let X, Y be random variables. The random variable X takes the values 1, 2 and 3 with probabilities $P(X=1) = 0,5$, $P(X=2) = 0,3$ & $P(X=3) = 0,2$,

& Y takes the values 1 & 2 with $P(Y=1) = 0,7$ & $P(Y=2) = 0,3$

Moreover, $P(X=1, Y=1) = 0,35$ & $P(X=3, Y=1) = 0,2$

a) Calculate the distribution funct. of X and Y

$$F_X(y) = \begin{cases} 0 & y < 1 \\ 0,5 & 1 \le y < 2 \\ 0,8 & 2 \le y < 3 \\ 1 & y \ge 3 \end{cases} \qquad F_Y(y) = \begin{cases} 0 & y < 1 \\ 0,7 & 1 \le y < 2 \\ 1 & y \ge 2 \end{cases}$$

b)

y \ x	1	2	3	
1	0,35	0,15	0,2	0,7
2	0,15	0,15	0	0,3
	0,5	0,3	0,2	

c) Independent?

No because:

$P(X=3, Y=2) = 0 \neq 0,3 \cdot 0,2$ ↯

Stochastics - KW3

(4)

Determine for which $a \in \mathbb{R}$ the function $f: \mathbb{R} \to \mathbb{R}$ with

$$f(x) = \begin{cases} ax & , x \in [0,1] \\ a(2-x) & , x \in [1,2] \\ 0 & , \text{else} \end{cases}$$

is a density and calculate $F_X(y)$

$f(x) \geq 0$ $\forall$ for $x \in [0,1]$ $\quad ax \geq 0$

for $a > 0$

- for $x \in (1,2]$ ~~$2a - ax \geq 0$~~

~~for~~ $a(2-x) \geq 0 \rightsquigarrow a \geq 0$

$\underbrace{\quad}_{\geq 0}$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_0^1 ax\,dx + \int_1^2 a(2-x)\,dx + 0$$

$$= \left[\frac{ax^2}{2}\right]_0^1 + \left[2ax - \frac{a}{2}x^2\right]_1^2$$

$$= \frac{a}{2} + 4a - a - (2a - 1) = \frac{a}{2} + 2a - 2a + \frac{a}{2} = \boxed{a = 1}$$

$\frac{a}{2} + \frac{a}{2}a - 4 + 1 - \frac{4}{2}a$ $\quad a = 1$

Statistics - KVB - 6xS

④

$$F_X(y) = \int_{-\infty}^{y} f(x)\,dx = \begin{cases} 0, & y < 0 \\ \in 0, & y > 2 \\ A, & 0 \le y \le 1 \\ B, & 1 < y \le 2 \end{cases}$$

$$A = \int_0^y a x\,dx = \left[\frac{a x^2}{2}\right]_0^y = \frac{a y^2}{2}$$

$$B = \int_0^1 a x\,dx + \int_1^y a(2-x)\,dx = \left[\frac{a x^2}{2}\right]_0^1 + \left[2ax - \frac{a}{2}x^2\right]_1^y$$

$$= \frac{-y^2}{2} + 2y - 1$$

$$C = \int_0^1 a x\,dx + \int_1^2 a(2-x)\,dx = 1$$

Stochastic Exercises

(5)

$$f_X(x) = \begin{cases} 1/2 & , x \in [2,4] \\ 0 & , \text{else} \end{cases}$$

a) Calculate $F_X(y)$

$$F_X(y) = \int_{-\infty}^{y} f(x)\,dx = \begin{cases} 0 & , y < 2 \\ A & , 2 \le y \le 4 \\ 1 & , y \ge 4 \end{cases}$$

$$A = \int_{-\infty}^{y} f(x)\,dx = \int_{2}^{y} \frac{1}{2}\,dx = \left[\frac{1}{2}x\right]_{2}^{y} = \frac{y}{2} - 1$$

b) Determine the density of the random variable $Y = 2X$

$$F_Y(y) = \begin{cases} 0 & , y/2 < 2 \\ \frac{y/2}{2} - 1 & , 2 \le y/2 \le 4 \\ 1 & , y/2 > 4 \end{cases} = \begin{cases} 0 & , x < 4 \\ \frac{1}{4}x - 1 & , 4 \le x \le 8 \\ 1 & , x > 8 \end{cases}$$

$$f_Y(y) = \begin{cases} \frac{1}{4}, & x \in [4,8] \\ 0, & \text{else} \end{cases}$$

Stochastics - K. 1 V 3

→ Expectation

Let (Ω, P) be a discrete probability space with a random variable X. The expectation $E[X]$ of X is defined by

$$E[X] = \sum_{\omega \in \Omega} X(\omega) \cdot P(\{\omega\})$$

if the sum on the right hand side is absolutely convergent.

The sum is absolutely convergent if $E[|X|] < \infty$

→ eg.:

For $\Omega = \{1, \dots, 6\}$, $P(A) = \frac{|A|}{6}$ for $A \subseteq \Omega$ and $X: \Omega \ni \omega \mapsto \omega$,

$$E[X] = \sum_{\omega \in \Omega} X(\omega) \cdot P(\{\omega\}) = \sum_{i=1}^{6} i \cdot \frac{1}{6} = \frac{7}{2}$$

→ For a random variable X with $E[|X|] < \infty$, we have

$$E[X] = \sum_{y \in X(\Omega)} y \cdot P(X = y)$$

→ eg.:

For $X = 1_A$ with $A \subseteq \Omega$

$$E[X] = 0 \cdot P(X=0) + 1 \cdot P(X=1) = P(1_A = 1) = P(A)$$

→ eg.: For $X \sim \text{Poisson}(\lambda)$ with $\lambda > 0$,

$$E[X] = \sum_{k=0}^{\infty} k \cdot P(X=k) = \sum_{k=0}^{\infty} k \cdot \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} e^{-\lambda}$$

$$= \sum_{k=0}^{\infty} \frac{\lambda^{k+1}}{k!} e^{-\lambda} = \lambda e^{-\lambda} \underbrace{\sum_{k=0}^{\infty} \frac{\lambda^k}{k!}}_{e^{+\lambda}} = \lambda$$

Stochastics - KW 8

3.36:

Let X & Y be random variables such that $E[|X|] < \infty$ and $E[|Y|] < \infty$. Then,

(i) $E[X+Y] = E[X] + E[Y]$
(ii) For $a \in \mathbb{R}$, $E[aX] = a \cdot E[X]$ } linear
(iii) If $X = C$ with $C \in \mathbb{R}$, then $E[X] = C$
(iv) If $X \geq 0$, then $E[X] \geq 0$
(v) If $X \geq Y$, then $E[X] \geq E[Y]$

Bsp:

A random experiment with success/failure and success probability is $p \in [0,1]$ is repeated n times.
Let X be the total number of successes.

Since $$X = \sum_{i=1}^{n} 1_{A_i}$$

with $A_i = \{\text{success in } i\text{-th repetition}\}$ for $i \in \{1,\dots,n\}$

$$E[X] = E\left[\sum_{i=1}^{n} 1_{A_i}\right] = \sum_{i=1}^{n} \underbrace{E[1_{A_i}]}_{P(A_i)=p} = n \cdot p$$

In general, for $Y \sim \text{Binomial}(n,p)$, $E[Y] = np$

Stochastik - KVB

→ 3.39:

Let X & Y be independent random variables such that $E[|X|]<\infty$ & $E[|Y|]<\infty$. Then, $E[|XY|]<\infty$ and

$$E[XY] = E[X]\,E[Y]$$

→ Expectation of general radon variables

Construction of expectation

Let $(\Omega, \mathcal{F}, P)$ be a probability space.

(i) For $X = \sum_{i=1}^{m} c_i \cdot 1_{A_i}$ with $c_1, \dots, c_m > 0$ and $A_1, \dots, A_m \in \mathcal{F}$ (such rV are called simple) let

$$E[X] = \sum_{i=1}^{m} c_i P(A_i)$$

(ii) For $X \geq 0$ construct simple random variables $X_n, n \in \mathbb{N}$ converging monotonously to X and let

$$E[X] = \lim_{n \to \infty} E[X_n]$$

(iii) For X define $X_+ = \max\{0, X\}$ and $X_- = \max\{0, -X\}$ or that $X = X_+ - X_-$ and define

$$E[X] = E[X_+] - E[X_-]$$

if $E[X_+] < \infty$ or $E[X_-] < \infty$

Stochastics - KIVB

→ 3.40:

For a random variable X with density f we have

$$E[X] = \int_{-\infty}^{\infty} x f(x)\, dx$$

if

$$\int_{-\infty}^{0} x f(x)\, dx < \infty \quad \text{or} \quad \int_{0}^{\infty} x f(x)\, dx < \infty$$

→ 3.41:

If

$$\int_{-\infty}^{\infty} |x| f(x)\, dx < \infty$$

we call X integrable and $E[X]$ is a finite number

→ eg:

For $X \sim \text{Uniformly}([a,b])$ with $a, b \in \mathbb{R}$, $a < b$

$$E[X] = \int_{a}^{b} x \frac{1}{b-a}\, dx = \left[\frac{x^2}{2(b-a)}\right]_a^b = \frac{a+b}{2}$$

→ eg:

For $X \sim \text{Exponential}(\lambda)$ with $\lambda > 0$ integration:

$$E[X] = \int_{0}^{\infty} x \lambda e^{-\lambda x}\, dx = [-x e^{-\lambda x}]_0^{\infty} - \int_0^{\infty} -e^{-\lambda x}\, dx$$

$$= 0 + \int_0^{\infty} e^{-\lambda x}\, dx = \left[-\frac{e^{-\lambda x}}{\lambda}\right]_0^{\infty} = \frac{1}{\lambda}$$

→ eg:

Let $N \sim Normal(0,1)$. Since

$$\int_0^\infty \frac{x}{\sqrt{2\pi}} e^{-x^2/2}\, dx = \left[\frac{-1}{\sqrt{2\pi}} e^{-x^2/2}\right]_0^\infty = \frac{1}{\sqrt{2\pi}} < \infty$$

we obtain

$$E[N] = \underline{\underline{0}}$$

Stochastics - KVB - Ex 6

① $X \sim \text{Binom}(n, p) \rightarrow P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$

$$E[X] = \sum_{k \in X(\Omega)} k P(X=k) = \sum_{k=0}^{n} k \cdot \binom{n}{k} p^k (1-p)^{n-k}$$

$$= \sum_{k=0}^{n} k \cdot \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

$$= \sum_{k=1}^{n} \frac{n!}{(k-1)!(n-k)!} p^k (1-p)^{n-k}$$

$$= \sum_{k=0}^{n-1} \frac{n \cdot (n-1)!}{k!(n-(k+1))!} p^{k+1} (1-p)^{n-(k+1)}$$

$$= n \cdot p \sum_{k=0}^{n-1} \frac{(n-1)!}{k!((n-1)-k)!} p^k (1-p)^{(n-1)-k}$$

$$= n \cdot p \cdot (p + 1 - p)^{n-1} = \underline{\underline{np}}$$

Stochastik - KA B - 6x6

(2)

a) $$f(x) = \begin{cases} x & ; x \in [0,1] \\ 2-x & ; x \in (1,2] \\ 0 & ; \text{else} \end{cases}$$

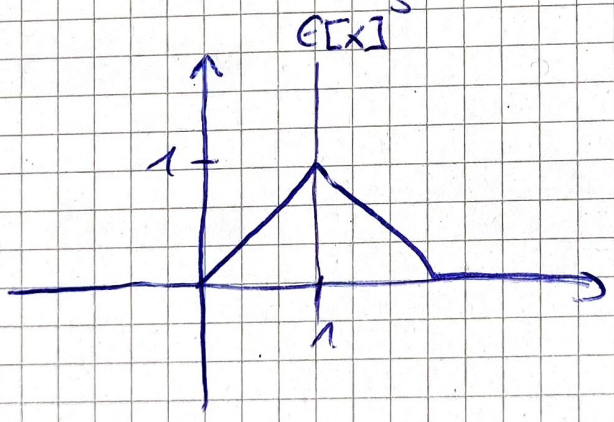
$$E[X] = \int_{-\infty}^{\infty} x f(x)\,dx = \int_0^1 x \cdot x\,dx + \int_1^2 x(2-x)\,dx + 0$$

$$= \left[\frac{x^3}{3}\right]_0^1 + \left[x^2 - \frac{x^3}{3}\right]_1^2$$

$$= \frac{1}{3} + 4 - \frac{8}{3} - \left(1 - \frac{1}{3}\right)$$

$$= -\frac{7}{3} + 4 - 1 + \frac{1}{3}$$

$$= -\frac{6}{3} + 4 - 1 = \underline{\underline{1}}$$



(2)

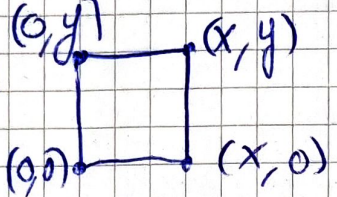
b) Let Y be a random variable distributed according to the Laplace distribution on the set $\{1,\dots,n\}$ for $n \in \mathbb{N}$.

$$P(Y=k) = \frac{1}{n}$$

$$E[Y] = \sum_{k=1}^{n} k \cdot \frac{1}{n} = \frac{1}{n} \sum_{k=1}^{n} k = \frac{n(n+1)}{n \cdot 2}$$

$$= \frac{n+1}{2}$$

c) R:



$R = X \cdot Y$

X & Y independent

$$E[XY] = E[X] \cdot E[Y] = \frac{n+1}{2}$$

Stochastic - KIB - 6x6

(3)

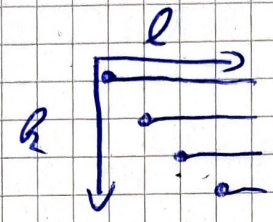
a) Let X denote a $\mathbb{N}_0$-valued random variable with finite expectation. Show that

$$E[X] = \sum_{k=1}^{\infty} P(X \geq k)$$

Idea:

$$\sum_{k=1}^{\infty} P(X \geq k) = P(X \geq 1) + P(X \geq 2) + \dots$$
$$= P(X=1) + 2\,P(X=2) + 3 \cdot P(X=3) + \dots$$

Proof:



$$\sum_{k=1}^{\infty} P(X \geq k) = \sum_{k=1}^{\infty} \sum_{\ell=k}^{\infty} P(X=\ell)$$
$$\overset{\ell \geq k}{=} \sum_{\ell=1}^{\infty} \sum_{k=1}^{\ell} P(X=\ell) = \sum_{\ell=1}^{\infty} P(X=\ell) \cdot \sum_{k=1}^{\ell} 1$$
$$= \sum_{\ell=1}^{\infty} P(X=\ell) \cdot \ell = E[X] \qquad \square$$

(3)

b) $X \sim$ geometrically distributed (p).
Show that $P(X \geq k) = p^k$ and compute $E[X]$

$$P(X = k) = p^k (1-p)$$

$$P(X \geq k) = \sum_{l=k}^{\infty} P(X = l) = \sum_{l=k}^{\infty} p^l (1-p)$$

$$= \sum_{l=0}^{\infty} p^{l+k} (1-p) = p^k \sum_{l=0}^{\infty} (1-p) p^l$$

$$= p^k \frac{(1-p)}{(1-p)} = p^k$$

$$E[X] = \sum_{k=1}^{\infty} p^k = \sum_{k=0}^{\infty} p^{k+1} = p \sum_{k=0}^{\infty} p^k = \frac{p}{1-p}$$

Stochastik - KVB - 6x6

④

For $n \in \mathbb{N}$ let $U_1, .., U_n$ denote independ and Vs which are uniformly distributed on $[0,1]$.
Furthermore, define Y as the maximum of $U_1, .., U_n$.
Show that the distributin function is given by

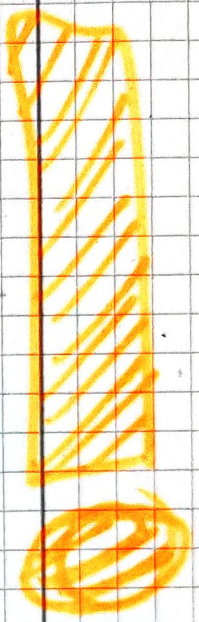
$$F_Y(x) = \begin{cases} 0 & , x < 0 \\ x^n & , x \in [0,1] \\ 1 & , \text{else} \end{cases}$$

$$Y = \left(\max_{k=1,..,n} U_1, .., U_n \right)$$

$$\Rightarrow P(Y \leq x) = P\left(\left[\max_{k=1,..,n} U_1, .., U_n \right) \leq x \right)$$

$$= P(U_1 \leq x, .., U_n \leq x) \overset{\text{indepth}}{=} P(U_1 \leq x)^n$$

$$= \begin{cases} 0 & , x < 0 \\ x^n & , x \in [0,1] \\ 1 & , \text{else} \end{cases} = F_Y(x)$$



$$f_Y(y) = \begin{cases} n y^{n-1} & , y \in [0,1] \\ 0 & , \text{else} \end{cases}$$

$$E[Y] = \int_{-\infty}^{\infty} y f_Y(y)\, dy = \int_0^1 y \cdot n \cdot y^{n-1}\, dy = \int_0^1 n \cdot y^n\, dy$$

$$= \left[\frac{n \cdot y^{n+1}}{n+1} \right]_0^1 = \frac{n}{n+1} \overset{n \to \infty}{\rightsquigarrow} 1$$

Stochastics - KVB

→ Functions of random variables

Let X be a random variable on a probability space $(\Omega, \mathcal{F}, P)$.

Then, for $A \in \mathcal{B}(\mathbb{R})$,

$$\{\omega \in \Omega : h(X(\omega)) \in A\} = \{\omega \in \Omega : X(\omega) \in \underbrace{h^{-1}(A)}_{\in \mathcal{B}(\mathbb{R})}\} \in \mathcal{F}$$

and $Y = h(X)$ is a random variable.

→ 3.45:

For a measureable $h: \mathbb{R} \to \mathbb{R}$ and a random variable X let $Y = h(X)$

(i) Let $X(\Omega)$ be at most countable. Then

$$E[Y] = \sum_{x \in X(\Omega)} h(x) P(X = x)$$

if

$$\sum_{x \in X(\Omega)} |h(x)| P(X = x) < \infty.$$

(ii) Let X have an absolutely continuous distribution (has density) with density f. Then

$$E[Y] = \int_{-\infty}^{\infty} h(x) f(x) \, dx, \text{ if } \int_{-\infty}^{\infty} |h(x)| f(x) \, dx < \infty$$

Stochastics - KVB

→ eg: 3.47:

For $X \sim \text{Poisson}(\lambda)$ with $\lambda > 0$,

$$E[X[X-1]] = \sum_{k=0}^{\infty} k(k-1) P(X=k) = \sum_{k=0}^{\infty} k(k-1) \frac{\lambda^k}{k!} e^{-\lambda}$$

$$= \sum_{k=2}^{\infty} k(k-1) \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k=2}^{\infty} \frac{\lambda^k}{(k-2)!} e^{-\lambda}$$

$$= \sum_{k=0}^{\infty} \frac{\lambda^{k+2}}{k!} e^{-\lambda} = \lambda^2 e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = \lambda^2 e^{-\lambda} e^{\lambda} = \lambda^2$$

→ eg: 3.48:

$$E[X^2] = \int_a^b x^2 \frac{1}{b-a} \, dx = \left[\frac{x^3}{3(b-a)} \right]_a^b$$

$$= \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 + ab + a^2}{3}$$

$X \sim \text{Uniform}([a,b])$

→ Variance:

Let X be a random variable such that $E[X^2] < \infty$.

Then variance of X is defined as

$$\text{Var}[X] = E[(X - E[X])^2]$$

→ eg:

Let X be a random variable such that

$P(X=i) = 1/6$ for $i \in \{1, \dots, 6\}$.

Since $E[X] = 7/2$, we obtain

$$\text{Var}[X] = \sum_{i=1}^{6} (i - 7/2)^2 \cdot P(X=i) = \sum_{i=1}^{6} (i - 7/2)^2 \frac{1}{6}$$

$$= \frac{1}{6} \left(\frac{25}{4} + \frac{9}{4} + \frac{1}{4} + \frac{1}{4} + \frac{9}{4} + \frac{25}{4} \right) = \frac{70}{24} = \frac{35}{12}$$

Stochastics - KVB

→ 3.51:

For a random variable X such that $E[X^2] < \infty$,

$$\mathrm{Var}[X] = E[X^2] - E[X]^2$$

→ eg: 3.52:

For $X \sim \mathrm{Poisson}(\lambda)$ with $\lambda > 0$,

$$\mathrm{Var}[X] = E[X^2] - E[X]^2 = E[X(X-1)+X] - E[X]^2$$
$$= E[X(X-1)] + E[X] - E[X]^2$$
$$= \lambda^2 + \lambda - \lambda^2 = \lambda$$

→ eg: 3.53:

For $X \sim \mathrm{Uniform}([a,b])$ with $a, b \in \mathbb{R}$, $a < b$,

$$\mathrm{Var}[X] = E[X^2] - E[X]^2$$
$$= \frac{b^2 + ab + a^2}{3} - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{12}$$

→ 3.54:

For a random variable X such that $E[X^2] < \infty$ and $a, c \in \mathbb{R}$,

$$\mathrm{Var}[aX] = a^2 \mathrm{Var}[X] \text{ and}$$
$$\mathrm{Var}[X + c] = \mathrm{Var}[X]$$

→ 3.56:

For a random variable X with $E[X^2] < \infty$, we denote $\sqrt{\mathrm{Var}[X]}$ as the standard deviation of X. This means that the parameter σ of the normal distribution is its standard deviation.

Stochastics - K & B

→ Covariance:

Let X and Y be random variables such that $E[X^2] < \infty$ and $E[Y^2] < \infty$. Then covariance of X and Y is defined as

$$\text{Cov}(X,Y) = E[(X - E[X]) \cdot (Y - E[Y])]$$

→ Lemma 3.58:

Let X and Y be random variables such that $E[X^2] < \infty$ and $E[Y^2] < \infty$.

(i) Then,
$$\text{Cov}(X,Y) = E[XY] - E[X]E[Y]$$

(ii) If X and Y are independent,
$$\text{Cov}(X,Y) = 0$$

$$\text{Cov}(a_1X_1 + a_2X_2, b_1Y_1 + b_2Y_2)$$
$$= a_1b_1\,\text{Cov}(X_1,Y_1) + a_2b_1\,\text{Cov}(X_2,Y_1)$$
$$+ a_1b_2\,\text{Cov}(X_1,Y_2) + a_2b_2\,\text{Cov}(X_2,Y_2)$$

Stochastik - KVO

→ Correlation / Coefficient:

Let X and Y be random variables such that $E[X^2] < \infty$ and $E[Y^2] < \infty$, $Var[X] > 0$ and $Var[Y] > 0$.

The correlation coefficient $\rho_{X,Y}$ of X and Y is defined as

$$\rho_{X,Y} = \frac{cov(X,Y)}{\sqrt{Var[X]} \cdot \sqrt{Var[Y]}}$$

and X and Y are called uncorrelated if $\rho_{X,Y} = 0$.

→ 3.62:

For random variables $X_1, .., X_n$ such that $E[X_1^2] < \infty, .., E[X_n^2] < \infty$

$$Var\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} Var[X_i] + 2 \sum_{1 \le i < j \le n} cov(X_i, X_j)$$

→ eg: 3.64

Let X be the total number of successes in n repetitions of a random experiment with success probability $p \in [0,1]$. Then,

$$X = \sum_{i=1}^{n} 1_{A_i} \quad \text{with } A_i = \{\text{success in } i\text{-th repetition}\} \text{ for } i \in \{1, .., n\} \text{ so that}$$

$$Var[X] = Var\left[\sum_{i=1}^{n} 1_{A_i}\right] = \sum_{i=1}^{n} Var[1_{A_i}] + 2 \sum_{1 \le i < j \le n} \underbrace{cov(1_{A_i}, 1_{A_j})}_{= 0 \text{ by Indepen.}}$$

$$= \sum_{i=1}^{n} Var[1_{A_i}] = n(p - p^2)$$

⇒ For $Y \sim Binomial((n,p))$: $Var[Y] = np(1-p)$

Stochastik KVB - 6x7

① Sei X ebenda a random variable with density

$$f(x) = \begin{cases} x & , X \in [0,1] \\ 2-x & , x \in (1,2] \\ 0 & , \text{else} \end{cases}$$

a) $E[X] = 1$.

$$\text{Var}[X] = E[x^2] - E[X]^2 = \frac{7}{6} - 1 = \frac{1}{6}$$

$$E[x^2] = \int_{-\infty}^{\infty} x^2 f(x)\,dx = \int_0^1 x^2 \cdot x\,dx + \int_1^2 x^2(2-x)\,dx$$

$$= \left[\frac{x^4}{4}\right]_0^1 + \left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_1^2 = \frac{7}{6}$$

Stochastics - KVB - 6x7

(2)

Let $n \in \mathbb{N}$ and X follow a discrete uniform distribution on $\{1, \dots, n\}$, i.e. $P(X = k) = 1/n$ for $k = 1, \dots, n$. Compute the variance of X.

$$E[X] = \frac{n+1}{2}$$

$$\mathrm{Var}[X] = E[X^2] - E[X]$$

$$E[X^2] = \sum_{k=1}^{n} k^2 P(X = k) = \sum_{k=1}^{n} k^2 \frac{1}{n} = \frac{1}{n} \sum_{k=1}^{n} k^2$$

$$= \frac{1}{n}\left(\frac{n^3}{3} + \frac{n^2}{2} + \frac{n}{6}\right) = \frac{n^2}{3} + \frac{n}{2} + \frac{1}{6}$$

(5)

Let $\lambda > 0$ and $X \sim \text{Poisson}(\lambda)$. Calculate $E[\lambda^{-X}]$

$\text{Poisson}(\lambda) \Rightarrow P(X=k) = \frac{\lambda^k}{k!} e^{-\lambda}$

$h(x) = \lambda^{-x}$

$$E[\lambda^{-X}] = \sum_{k=0}^{\infty} \lambda^{-k} \frac{\lambda^k}{k!} e^{-\lambda}$$

$$= \boxed{\sum_{k=0}^{\infty} \frac{1}{k!}} \cdot e^{-\lambda} = e \cdot e^{-\lambda} = \frac{e}{e^{\lambda}}$$

$\boxed{\ldots} = e$

Stochastik - KVB - E<7

(9)

Betrachte zwei Zufallsvariablen X & Y, die erfüllen $Var[X] = 2$, $Var[Y] = 3$ und $\rho_{X,Y} = 0,5$.
Berechne $Var[3X - 2Y]$ & $Cov(3X + Y, 2X - 5Y)$

$$\rho_{X,Y} = \frac{Cov(X,Y)}{\sqrt{Var[X]}\sqrt{Var[Y]}} \Leftrightarrow 0,5 = \frac{Cov(X,Y)}{\sqrt{2}\cdot\sqrt{3}}$$

$$\Leftrightarrow Cov(X,Y) = \frac{\sqrt{6}}{2}$$

$$Cov(3X+Y, 2X-5Y) = 3\cdot 2 \cdot \underbrace{Cov(X,X)}_{=Var[X]} + 1\cdot 2\, Cov(X,Y) + 3\cdot(-5)\cdot Cov(X,Y) + 1\cdot(-5)\underbrace{Cov(Y,Y)}_{Var[Y]}$$

$$= -18,9$$

$$Var[3X - 2Y] = Cov(3X - 2Y, 3X - 2Y)$$

$$= 9\,Var[X] - 12\,Cov(X,Y) + 4\,Var[Y] = 19,3$$

Stochastik - KVS - Ex 7

(5)

def $\alpha > 0$ and $X \sim \text{Pareto}(\alpha)$.

$$X \sim \text{Pareto}(\alpha) \Rightarrow f(x) = \begin{cases} 0 & , x < 1 \\ \alpha x^{-\alpha-1} & , x \geq 1 \end{cases}$$

a) Compute $E[X]$.

$$E[X] = \int_{-\infty}^{\infty} x f(x) = \int_1^{\infty} x \cdot \alpha \cdot x^{-\alpha-1} \, dx = \int_1^{\infty} \alpha \cdot x^{-\alpha} \, dx$$

$$= \left[\frac{\alpha x^{-\alpha+1}}{-\alpha+1} \right]_1^{\infty}$$

$$= \frac{\alpha (\infty)^{-\alpha+1}}{-\alpha+1} - \frac{\alpha}{-\alpha+1}$$

for $\alpha > 1$: $E[X] = \frac{\alpha}{\alpha - 1}$

b) $E[X^2]$, $\text{Var}[X]$

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x) \, dx = \int_1^{\infty} x^2 \alpha x^{-\alpha-1} \, dx = \int_1^{\infty} \alpha x^{-\alpha+1} \, dx$$

$$= \left[\frac{\alpha x^{-\alpha+2}}{-\alpha+2} \right]_1^{\infty} = \frac{\alpha (\infty)^{-\alpha+2}}{-\alpha+2} + \frac{\alpha}{\alpha - 2}$$

$\alpha > 2$

$$= \frac{\alpha}{\alpha - 2}$$

$$\text{Var}[X] = \frac{\alpha}{\alpha-2} - \frac{\alpha}{\alpha-1} = \frac{\alpha}{(\alpha-2)(\alpha-1)^2}, \text{ if } \alpha > 2$$

(5)

c) $f_Y(y) = \begin{cases} \frac{1}{2}\alpha |y|^{-\alpha-1} & , \text{für } |y| \geq 1 \\ 0 & , \text{else} \end{cases}$

3.40: $\int_{-\infty}^{0} x f_Y(x)\,dx < \infty$ or $\int_{0}^{\infty} x f_Y(x)\,dx < \infty$

$$\rightarrow \int_{0}^{\infty} x f_Y(x) = \int_{1}^{\infty} \frac{1}{2} x \alpha x^{-\alpha-1}\,dx = \frac{1}{2} E[X] = \begin{cases} < \infty & , \alpha > 1 \\ = \infty & , \alpha \leq 1 \end{cases}$$

$$\rightarrow -\int_{-\infty}^{0} x f_Y(x)\,dx = -\int_{-\infty}^{-1} x \cdot \frac{1}{2} |x|^{-\alpha-1}\,dx = \frac{1}{2}\int_{-\infty}^{-1} -x \alpha |x|^{-\alpha-1}\,dx$$

$$= \frac{1}{2}\int_{1}^{\infty} x \alpha x^{-\alpha-1}\,dx = \frac{1}{2} E[X] = \begin{cases} < \infty & , \alpha > 1 \\ = \infty & , \alpha \leq 1 \end{cases}$$

$\rightsquigarrow \alpha > 1$!

$$E[Y] = \int_{-\infty}^{\infty} x f_Y(x)\,dx = \int_{-\infty}^{0} x f_Y(x)\,dx + \int_{0}^{\infty} x f_Y(x)\,dx$$

$$= -\left(-\int_{-\infty}^{0} x f_{-Y}(x)\,dx\right) + \int_{0}^{\infty} x f_{-Y}(x)\,dx$$

$$= -\frac{1}{2} E[X] + \frac{1}{2} E[X] = \underline{\underline{0}} \quad , \alpha > 1$$